



DABBAL SOFTWARE DEVELOPMENT PLC

A PLACE FOR YOUR TECHNOLOGICAL NEED

Proposal Document

**For Development of Advanced Queue
Management System (QMS) +
Performance Management System (PMS)**

Prepared by:
Dabbal Software Development PLC

For:
Amente Electronics PLC

Date:
August 19, 2025

1. Executive Summary

Amente Electronics PLC has expressed the need for an **enterprise-grade Queue Management System (QMS)** integrated with a **Performance Management System (PMS)** to enhance customer service delivery, optimize staff performance, and scale operations to handle **up to 130 departments**, a significant upgrade from the current **16-department capacity**.

This proposal outlines the **system features, scalability enhancements, performance improvements, and reporting mechanisms** required to deliver a **robust, future-proof solution**.

2. Project Objectives

- Develop a **Next-Generation QMS + PMS** to streamline queue handling and staff evaluation.
 - Ensure **horizontal and vertical scalability** to support up to **130 departments simultaneously**.
 - Provide **real-time insights, performance tracking, and reporting** for effective management.
 - Deliver a **secure, compliant, and optimized system** that can integrate with existing enterprise applications (ERP/CRM).
-

3. System Features & Modules

3.1 Advanced Queue Management Features

Queue Management Module

- Walk-in Kiosk (touchscreen with QR code ticketing).
- Enhancing and Scaling horizontally & vertically to 130 departments.
- Pending Ticket management (hold & recall function).
- Delay Management (pending request handling).
- Extended system optimization to handle **130+ simultaneous windows (Fodda)** with stable performance.

Smart Queue Routing Module

- Semi-AI-based load balancing: shortest wait time assignment.
- Service-based prioritization: VIP, elderly, emergency support.
- Dynamic re-routing when counters/windows are overloaded.

Real-Time Customer Engagement Module

- Voice + Screen display calling system (TV & speakers).
 - Multi-language voice announcements (Affan Oromo, Amharic, English, etc.).
-

3.2 Advanced Performance & Evaluation Module

Staff Metrics Module

- Track Average Handling Time, First Response Time, Ticket Completion Rate.

Staff Evaluation Dashboard

- Real-time staff KPIs.
- Leaderboards & comparison analytics.

HR Integration

- Performance-linked reward and evaluation system.
 - Monthly auto-reports for HR decision-making.
-

3.3 Reporting & Analytics Module

Operational Reports

- Daily/Weekly/Monthly trends.
- Peak-hour analysis (heatmaps).
- Wait time vs service time analytics.
- Abandonment rate tracking.

Business Intelligence Reports

- Demand forecasting.
- Branch/Window benchmarking.

Custom Reports

- Customizable dashboards.
 - Export to Excel/PDF/CSV.
-

3.4 Security & Compliance

- Enhanced role-based access control (Admin, Supervisor, Staff, Viewer).
 - Data privacy compliance with secured logs.
 - System-wide audit trail for accountability.
 - Secure logins with encrypted authentication.
-

3.5 Scalability & Multi-Department Support

- Centralized cloud dashboard for HQ oversight.
 - Performance comparison across **up to 130 departments**.
 - Unified reporting for regional/national visibility.
 - API integrations with ERP/CRM systems.
-

3.6 Feedback Evaluation

- On-premise or kiosk-based customer feedback tablets.
 - Real-time service satisfaction rating.
-

3.7 Advanced KPI Dashboard

- Customer Queue Overview: live loads & estimated wait times.
 - Staff Leaderboard: efficiency rankings.
 - Service Demand Analysis: real-time service popularity.
 - Window Comparison: inter-department performance.
 - Customer Satisfaction Index (CSI): Net Promoter Score (NPS) in real-time.
-

4. System Enhancement for Scaling

To upgrade from **16 departments** → **130 departments**, the following enhancements are required:

- **Database Optimization:**
High-performance relational database with indexing & partitioning for scalability.
- **Load Balancing & Performance Tuning:**
Multi-server cluster architecture with horizontal scaling support.

- **Infrastructure Upgrade:**
Cloud-native or hybrid deployment with microservices architecture.
 - **Parallel Processing Enhancements:**
Multi-threaded queue handling for concurrent requests.
 - **Resiliency & Fault Tolerance:**
Failover servers, auto-recovery, and redundancy mechanisms.
 - **Extended Analytics Engine:**
Capable of processing **high-volume real-time data** for 130+ departments simultaneously.
-

5. Implementation Timeline

Phase	Activities	Duration
Phase 1	Requirement Analysis & System Design	1 day
Phase 2	Development of Core Modules (Queue, Performance, Reporting)	14 days
Phase 3	Scaling Enhancements & Optimization	7 days
Phase 4	Integration, Testing & QA	5 days
Phase 5	Deployment & Training	3 day
Total		30 days

6. Investment & Cost Estimate

Component	%	Subtotal (ETB)
Design & Requirements	7%	210,000
Frontend Development	27%	810,000
Backend & API Development	40%	1,200,000
Integration & Testing	17%	510,000
Deployment & Handover	9%	270,000
Total	100%	3,000,000
VAT (10%)		300,000
Grand Total		3,300,000 ETB

7. Deliverables

- Fully functional **Advanced QMS + PMS system**.
 - System documentation (technical + user manuals).
 - Training sessions for staff & administrators.
 - Post-deployment **support & maintenance**.
-

8. Conclusion

This solution empowers **the company** with a **future-proof, scalable, and data-driven system** that not only manages queues efficiently but also **evaluates staff performance, enhances customer satisfaction, and supports up to 130 departments**.

Dabbal Software Development PLC commits to delivering a **robust, secure, and scalable enterprise solution** that supports both current needs and future growth.